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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Artificial Intelligence and Data Science

Academic Year: 2024- 2025 (ODD Semester)

Active Learning Best practices: Flipped Classroom – User-Defined Routines & Collection Types

Degree, Semester & Branch: III Sem. B.Tech. Artificial Intelligence and Data Science – 'A'

Course Code & Title: - AD3391 – Database Design and Management

Name of the Faculty member: Mrs. G. Kavitha, AP/AI & DS

Theme of discussion: Flipped Classroom

Topics Covered: Unit- V - User-Defined Routines & Collection Types

Date and Time: 09.11.2024 & 04.30 P.M to 05.15 P.M

Active Learning Best practices: Flipped Classroom – User-Defined Routines & Collection Types

Unit / Topic: Unit – V / User-Defined Routines & Collection Types

Course Outcome: CO5 - Apply the data model and querying in Object-relational and NoSQL databases.

Topic Learning Outcome: TLO47

Activity Chosen: Flipped Classroom

Justification:

- The concept of user-defined routines & collection types lecture was posted into the student's group, to identify the learning level of students about the user-defined routines & collection types activity was given. It is an important topic. The question was asked in most of the university questions.

Time Allotted for the Activity: 30 Minutes

Details of the Implementation:

The flipped classroom activity was implemented as a teaching strategy to enhance student engagement, foster deeper understanding, and promote active learning. In a traditional classroom, students typically receive lectures during class time and do homework or assignments outside of the class. However, in a flipped classroom, this model is reversed - students watch pre-recorded lectures or read material at home and on the day of activity ask the students to explain about the concept. This report summarizes the implementation and outcomes of the flipped classroom activity.

User-Defined Routines & Collection Types:

User-Defined Routines (UDRs)

UDRs are custom routines created by users to encapsulate complex logic or operations that are not directly supported by SQL's built-in functions. These routines are typically used for tasks like data validation, custom calculations, or complex data transformations.

Types of UDRs:

1. Stored Procedures:
   - A block of SQL statements saved in the database that can be executed as a single unit.
   - Often used for performing a sequence of operations like data manipulation and validation.
   - Can include control-flow constructs like loops and conditionals.

Example:

```
CREATE PROCEDURE UpdateEmployeeSalary (emp_id INT, new_salary DECIMAL)
BEGIN
  UPDATE Employees
  SET Salary = new_salary
  WHERE EmployeeID = emp_id;
END;
```

2. User-Defined Functions (UDFs):
   - Functions that return a value or a table.
   - Often used in SELECT statements for performing calculations or transformations.

Example (Scalar Function):

```
CREATE FUNCTION GetFullName (first_name VARCHAR(50), last_name VARCHAR(50))
RETURNS VARCHAR(100)
BEGIN
  RETURN CONCAT(first_name, ' ', last_name);
END;
```

Example (Table-Valued Function):

```
CREATE FUNCTION GetActiveEmployees()
RETURNS TABLE
AS
RETURN
(
  SELECT * FROM Employees WHERE Status = 'Active'
);
```

3. Triggers:
   - A special type of UDR that is automatically executed in response to certain events on a table, like INSERT, UPDATE, or DELETE.

Example:

```
CREATE TRIGGER trg_AfterInsert ON Orders
AFTER INSERT
AS
BEGIN
  PRINT 'A new order has been added!';
END;
```

Advantages of UDRs:

- Promote code reuse.
- Improve maintainability by centralizing complex logic.
- Enhance database security by limiting direct access to tables.

Collection Types

Collection types in SQL refer to data types that store multiple values in a single variable or column. They are particularly useful for representing complex data structures or handling data with an unknown number of elements.

Common Collection Types:

1. Arrays:
   - Store a fixed or variable number of elements of the same data type.
   - Supported in some databases like PostgreSQL.

Example:

```
CREATE TABLE Products (
  ProductID SERIAL PRIMARY KEY,
  Tags TEXT[]
);

INSERT INTO Products (Tags) VALUES (ARRAY['Electronics', 'Gadgets']);
```

2. Nested Tables (PL/SQL):
   - A collection type that behaves like a single-column table that can be nested within another table.
   - Commonly used in Oracle databases.

Example:

```
DECLARE
  TYPE NameList IS TABLE OF VARCHAR2(50);
  names NameList := NameList('Alice', 'Bob', 'Charlie');
BEGIN
  FOR i IN 1..names.COUNT LOOP
    DBMS_OUTPUT.PUT_LINE(names(i));
  END LOOP;
END;
```

3. VARRAYs (Variable-Sized Arrays):
   - A fixed-size array in PL/SQL, allowing multiple elements of the same type.

Example:

```
CREATE TYPE PhoneList AS VARRAY(5) OF VARCHAR2(15);
```

4. JSON and XML Collections:
   - JSON and XML data types can act as collections, holding arrays or nested objects.
   - Supported by databases like PostgreSQL, MySQL, and SQL Server.

Example (JSON in PostgreSQL):
```
CREATE TABLE Employees (
   EmployeeID SERIAL PRIMARY KEY,
   Details JSONB
);

INSERT INTO Employees (Details)
VALUES ('{"Name": "John Doe", "Skills": ["SQL", "Java"]}');
```

5. Associative Arrays (PL/SQL):
   - Key-value pairs indexed by integers or strings.
   - Flexible and efficient for lookup operations.

Example:
```
DECLARE
   TYPE AssocArray IS TABLE OF VARCHAR2(50) INDEX BY PLS_INTEGER;
   emp_names AssocArray;
BEGIN
   emp_names(1) := 'Alice';
   emp_names(2) := 'Bob';
   DBMS_OUTPUT.PUT_LINE(emp_names(1));
END;
```
Advantages of Collection Types:
- Simplify the management of related data.
- Reduce the need for complex table joins.
- Enable more dynamic data processing within applications.

CO – PO / PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	2	2	2	2	2	2	2	2	3	2	3

PO/PSO Mapped

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO9
	3	2	2	2	2	2
Justification for correlation	Students can be able to apply the concepts of user defined routines on object-relational databases to solve engineering problems.	Students can be able to analyze the usage of user defined routines.	Students can be able to design efficient queries and apply advanced data types for developing effective database solutions.	Students can be able to analyse how different data types affect performance and storage efficiency.	Students can be able to perform user defined operations in SQL by using modern database tools and technologies.	Students can be able to use user defined routines in complex database solutions involving teamwork and communication

Innovative Practice	PO10	PO11	PO12	PSO1	PSO2	PSO3
	2	2	2	3	2	3
Justification for correlation	Students can be able to communicate user defined routines concepts to diverse stakeholders, including technical and non-technical stakeholders.	Students can be able to analyze the implications of different collection types and their applications.	Students can be able to update with new techniques and tools.	Students can be to apply user defined routines & collection types concepts and techniques to solve a variety of engineering problems.	Students can be able to user defined routines & collection types to support real-time analytics.	Students can be able to apply user defined routines & collection types in applications that forecast future trends.

Glimpses:

Reflective Critique:

- Preparing and Sharing Content: Video lectures and reading materials were created covering the relevant topics. These materials were shared with the students through the learning management system or other suitable platforms well in advance of the corresponding class sessions.
- Class Sessions: During the class sessions, students were asked to discuss and apply the concepts covered in the pre-recorded lectures. Facilitators were present to guide and support the students' learning process.
- Active Learning Activities: Students are asked to explain about the concept in front of the classroom, and peer teaching sessions were conducted.
- Feedback Collection: Students were asked to provide feedback on the flipped classroom experience, highlighting the aspects they found beneficial and suggesting potential improvements.

Observations:

The implementation of the flipped classroom activity was successful in enhancing student engagement, promoting deeper understanding, and creating a more interactive learning environment. By using this approach, students were encouraged to take ownership of their learning process and actively participate in class activities. However, continuous refinement based on student feedback and ongoing assessment of the approach will be essential to optimize its effectiveness in future implementations.

Benefit of the practice:

- The students understand well about use of user-defined routines & collection types.
- The students can able to interact with the faculty member and their classmates during the class was effective.
- It helps to improve the self-learning of a student.
- With the help of this activity, each student can get clear idea about the use of user-defined routines & collection types in object relational databases.

Students Response:

- Bright students were actively participated to prepare the topic wise questions and concept notes.
- Slow learners expect to assist the topic towards searching online and also understanding the questions
- Most of the students were enjoyed the session.
- Students response ensures the students could improve listening, communication, creativityand problem-solving skills.

12/11/24

Signature of the Faculty Member

12/11

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Department of Artificial Intelligence and Data Science

Academic Year: 2024- 2025 (ODD Semester)

Active Learning Best practices: Flipped Classroom – User-Defined Routines & Collection Types

Degree, Semester & Branch: III Sem. B.Tech. Artificial Intelligence and Data Science – 'A'

Course Code & Title: - AD3391 – Database Design and Management

Name of the Faculty member: Mrs. G. Kavitha, AP/AI & DS

Theme of discussion: Flipped Classroom

Topics Covered: Unit- V - User-Defined Routines & Collection Types

Date and Time: 11.11.2024 & 05.15 P.M to 06.00 P.M

Feedback Questions:

1. Did the plan align with your expectations?

   Yes No

2. How effectively was the innovative practice plan implemented?

   Excellent Good Satisfactory

3. How well did you understand the objectives and purpose of the innovative practice plan?

   1 2 3 4 5

4. Were the necessary resources and support provided for successful implementation?

   Yes No

5. Did you notice any specific benefits or changes resulting from the plan?

   Yes No

6. How engaged were you throughout the practice plan?

   1 2 3 4 5

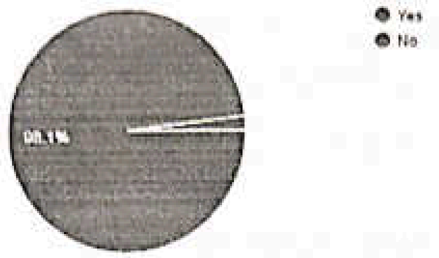
7. Did the plan provide opportunities for active participation and collaboration?

   Yes No

Feedback Analysis:

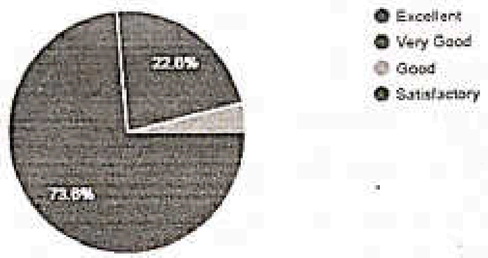
Did the plan align with your expectations? Copy

53 responses



How effectively was the innovative practice plan implemented? Copy

53 responses



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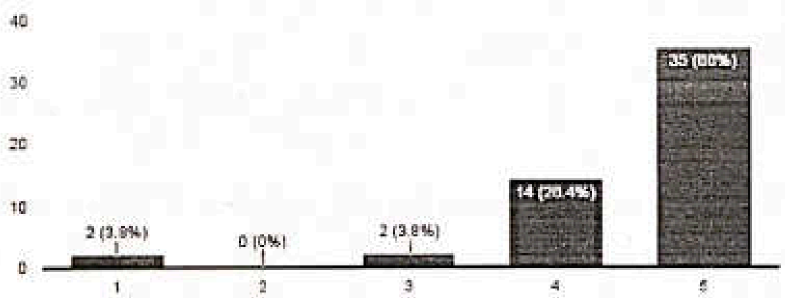

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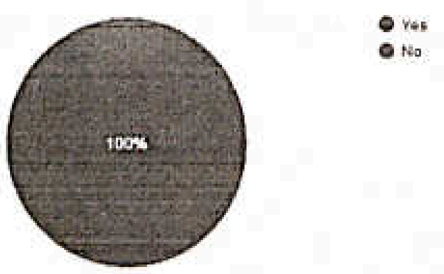

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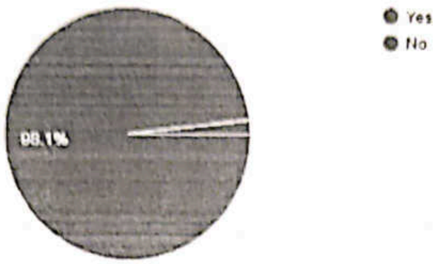

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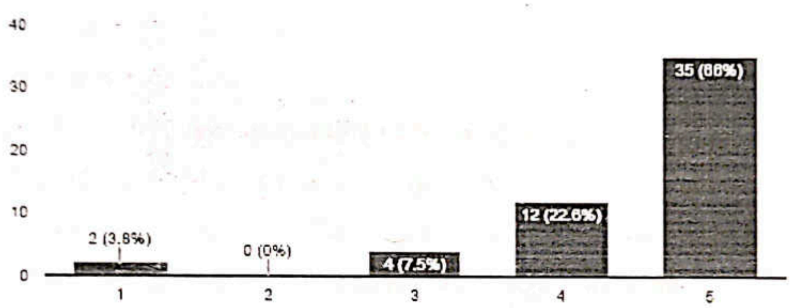

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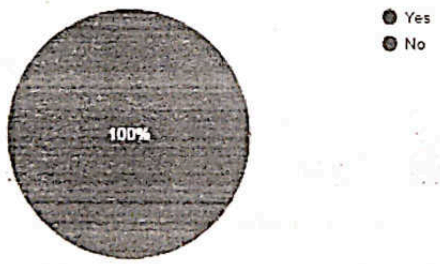

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Academic Year: 2024- 2025 (ODD Semester)

Active Learning Best practices: Flipped Classroom – User-Defined Routines & Collection Types

Degree, Semester & Branch: III Sem. B.Tech. Artificial Intelligence and Data Science – 'B'

Course Code & Title: - AD3391 – Database Design and Management

Name of the Faculty member: Mrs. G. Kavitha, AP/AI & DS

Theme of discussion: Flipped Classroom

Topics Covered: Unit- V - User-Defined Routines & Collection Types

Date and Time: 11.11.2024 & 05.15 P.M to 06.00 P.M

Active Learning Best practices: Flipped Classroom – User-Defined Routines & Collection Types

Unit / Topic: Unit – V / User-Defined Routines & Collection Types

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CO – PO / PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
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PO/PSO Mapped

Innovative Practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO9**
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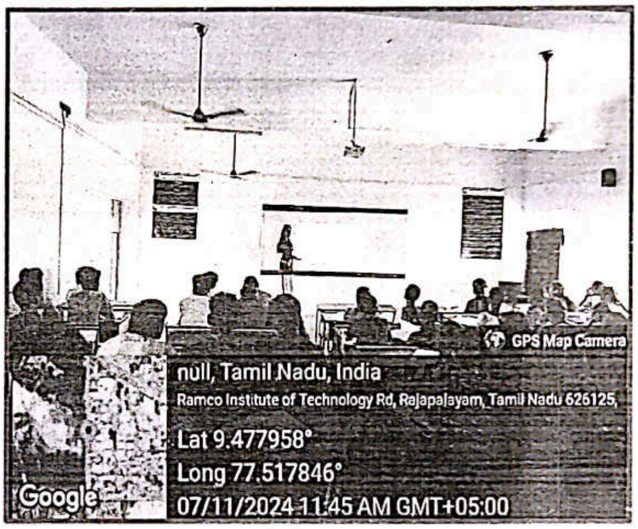
Glimpses:



Reflective Critique:

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Feedback Questions:

1. Did the plan align with your expectations?

   Yes No

2. How effectively was the innovative practice plan implemented?

   Excellent Good Satisfactory

3. How well did you understand the objectives and purpose of the innovative practice plan?

   1 2 3 4 5

4. Were the necessary resources and support provided for successful implementation?

   Yes No

5. Did you notice any specific benefits or changes resulting from the plan?

   Yes No

6. How engaged were you throughout the practice plan?

   1 2 3 4 5

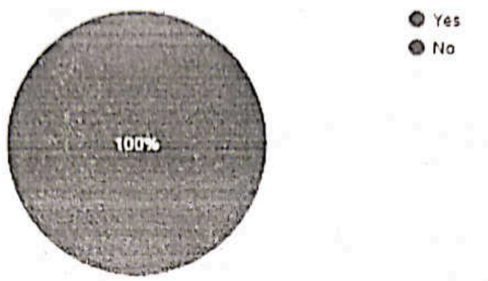
7. Did the plan provide opportunities for active participation and collaboration?

   Yes No

Feedback Analysis:

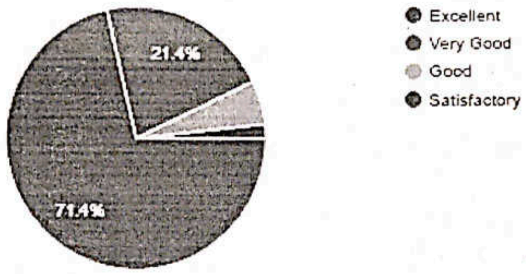
Did the plan align with your expectations? Copy

56 responses



How effectively was the innovative practice plan implemented? Copy

56 responses



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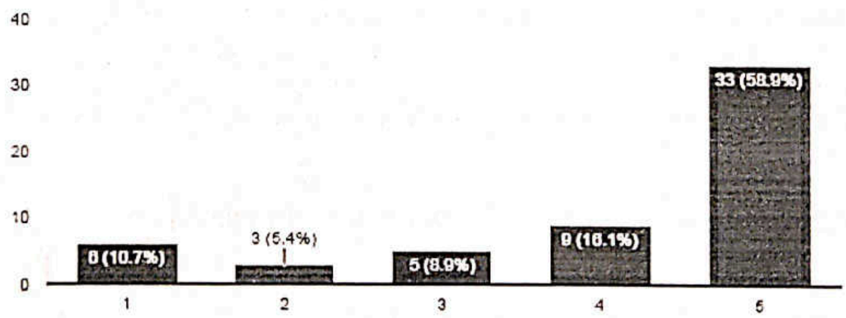

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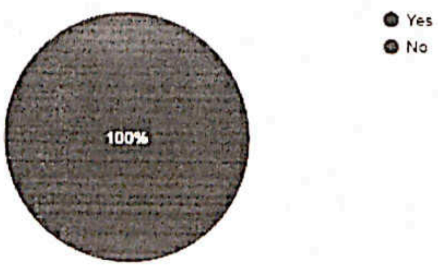

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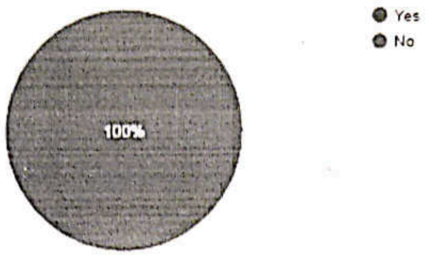

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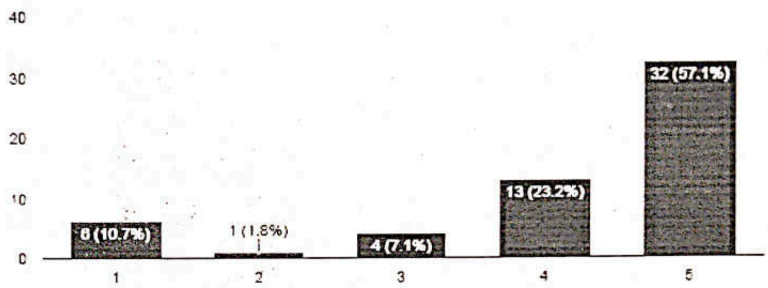

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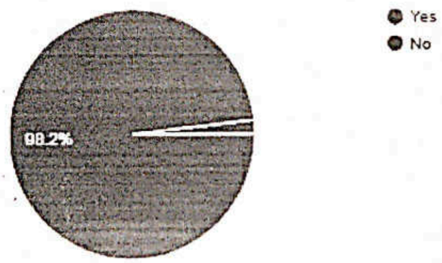

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